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Mechanical stability of Scots pine, Norway spruce and birch: an analysis of tree-pulling experiments in Finland

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Abstract

Tree pulling experiments were conducted in Eastern Finland (1995–1996) involving 51 Scots pines (*Pinus sylvestris* L.), 33 Norway spruces (*Picea abies* (L.) Karst.) and 11 birches (*Betula* spp.) under unfrozen soil conditions and 20 Scots pines when the (podzol) soil was frozen. The trees were pulled over with a winch attached at a stem height of 6 m and the force required to uproot or break the tree was recorded, together with detailed measurements of the physical characteristics of the tree. Analysis of the data showed that the maximum resistive bending moment was most significantly and positively correlated with diameter at breast height (DBH) and tree height (*H*). Tree height multiplied by the second power of DBH provided the best prediction of the maximum resistive bending moment required for uprooting, allowing the species to be ranked in the following decreasing order of resistance: Scots pine>birch>Norway spruce. Correspondingly, the third power of breast height diameter served best to explain the maximum resistive bending moment for stem breakage, with the following decreasing order of resistance: birch>Scots pine>Norway spruce. Highly tapering trees were more liable to stem breakage than uprooting, and vice versa. All the Scots pines broke under frozen soil conditions, and higher bending moments were needed for stem breakage than under unfrozen soil conditions. The modulus of rupture of the broken stems, as derived from the maximum resistive bending moments for stem breakage, showed the following decreasing order of resistance: birch>Scots pine>Norway spruce.
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Keywords: Tree pulling; Winch system; Maximum resistive bending moment; Modulus of rupture; Wind damage

1. Introduction

Forest damage by wind and snow is a continual cause of economic loss in managed forests, where tree instability imposes important restrictions on silviculture, especially at sites where there is a high risk of damage. Wind and snow damage results in loss of timber yield, the economic impact being particularly severe in semi-mature forests that are growing rapidly. This is because each year of additional growth can bring substantial increases in total timber volume, tree size and value. Both, wind and snow damage will reduce the yield of recoverable timber and increase the costs of unscheduled thinnings leading to disturbances in planned forestry management. Furthermore, broken and uprooted trees in stands can lead to detrimental insect attacks on the remaining stems because of the increased availability of breeding material (Ravn, 1985; Schroeder and Eidmann, 1993).

A number of approaches have been adopted to date in the study of the stability of forests, including field experiments on the effects of thinning and fertilization (Laiho, 1987; Lohmander and Helles, 1987; Persson,

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1972, 1975; Cremer et al., 1982), measurements of wind and the response of trees to it (Oliver and Mayhead, 1974; Papesch, 1974; Gardiner, 1994; Peltola, 1996), wind tunnel measurements predicting the stability of stands (Fraser, 1962; Walshe and Fraser, 1963; Stacey et al., 1994; Gardiner et al., 1997) and statistical studies involving tree pulling, in which the force required to pull a tree over using a winch is related to the physical characteristics of the tree, forest treatment and soil type (Fraser, 1962; Fraser and Gardiner, 1967; Somerville, 1979; Smith et al., 1987; Coutts, 1986; Papesch et al., 1997; Ray and Nicoll, 1998; Moore, 1999). The results of this research suggest that stand and ground characteristics such as tree species, tree dimensions, stand density, soil type, determine the critical wind speed and/or snow load at which damage will occur (e.g. Persson, 1972, 1975; Rottmann, 1985; Lohmander and Helles, 1987; Valinger et al., 1993; Solantie, 1994; Nykänen et al., 1997), whereas the local wind and snow extremes and the topography will determine how probable such a wind speed and/or snow load is.

Based on the above understanding of tree response, attempts have been made to predict the risk of wind and snow damage to trees, i.e. to understand the effect of forces acting on trees (Petty and Worrell, 1981; Petty and Swain, 1985; Peltola and Kellomäki, 1993; Peltola et al., 1999). Mechanistic models (e.g. Peltola and Kellomäki, 1993; Peltola et al., 1999) have recently been under development for predicting the critical wind speeds at which trees are likely to be uprooted or broken; i.e. to provide tools for assessing the risk of wind and snow damage in the context of forest management. However, much basic work is still needed, especially with regard to the components of root anchorage (because of the complexity of the root-soil system), and also with regard to stem stability. These can be investigated using static loads, with the reservation that the results may need to be modified when the dynamic forces caused by wind are introduced (Coutts, 1986).

In a static system the uprooting forces, usually calculated as bending moments at the base of the stem, are treated as arising in two ways. Firstly, the force produced by wind action on the crown, simulated by pulling with a rope, causes deflection of the stem. The leaning stem then assists in uprooting the tree because its centre of gravity moves over the hinge point in the root system (Ray and Nicoll, 1998). Thus, a second uprooting force is provided by the weight of the stem and crown. The uprooting moment is resisted by bending of the tree stem and various components of root anchorage: the weight of the root-soil plate, the strength of the windward roots, the strength of the root hinge and the soil strength at the base of the root-soil plate. If the uprooting moment exceeds the resistive bending moment of the tree at a particular angle of deflection, the tree will deflect further. The tree will give way if the uprooting moment exceeds its maximum resistive bending moment, with the relative strengths of the stem and roots determining the mode of failure (Petty and Worrell, 1981).

We report here on tree-pulling experiments designed to determine the relationships between the maximum resistive bending moment of a tree, which quantifies its ability to resist an applied load, and various measures of its size. The experiments involved pulling Scots pines, Norway spruces and birches over on podzolic soils under unfrozen soil conditions and Scots pines under frozen soil conditions. The latter experiments were performed especially to find out whether any difference exists in the maximum resistive moment needed to cause stem breakage between frozen and unfrozen soil conditions.

2. Material and methods

2.1. Site and stand information

The tree-pulling experiments were carried out in 1995–1996 in stands of Scots pine (*Pinus sylvestris* L.), Norway spruce (*Picea abies* (*Karst*) K.) and birch (*Betula* spp.) growing on podzol soils in Eastern Finland (Table 1). The sites were of medium fertility, representing the *Myrtillus* (stands 2–4) and *Vaccinium* (stand 1) types, and thus possessed the most appropriate soils for the growth of pine, spruce and birch in Finland.

2.2. Tree-pulling tests

2.2.1. Measurement of maximum resistive bending moment

A winch system was used to pull the trees and the maximum applied force needed to uproot a tree or

Table 1
Stand informatiossn

Stand	Soil type	Soil density (kg/m^2)	Tree species (%)	Stand age (years)	Stand density (stems ha^{-1})	DBH (cm)	Height (m)
1	Medium and fine silt	1808	Scots pine (100)	100	1300	20	16
2	Fine sand and course silt	1210	Scots pine (10), Norway spruce (82), Birch sp (8)	80	1600	16	16
3	Medium and fine silt	1395	Scots pine (100)	40	1800	14	13
4	Medium and fine silt	1894	Norway spruce (100)	80	1000	23	20

break its stem was measured at the base of the stem (Fig. 1). Depending on the size of the tree, either a hand winch was used (pulling force of up to 40 kN) or a Valmet Terra tractor with a winch system (pulling force of up to 80 kN with double rope). The pulling force was measured by means of a load cell (Philips PR6206/53N) connected to a digital weighing unit (Philips Digital KS380) which could measure loads of up to 5000 kg.

The attachment point was at a constant height of 6 m above the ground, representing from one-third to one-half of the tree height. This height, mostly under or near the crown base, enabled the modulus of elasticity (MOE) of the stem to be measured at the same time, too. The attachment height used was in line with those ones used in many previous studies (Fraser, 1962; Fraser and Gardiner, 1967; Smith et al., 1987; Frederickson et al., 1993; Moore, 1999). But, it would not yield a uniform stress profile in the outer fibres, because in this case the height should be ≈80% (Wood, 1995). In this study, trees pulled from below this point would, thus, obviously have the largest stresses, and hence the break point in the lower portion of the stem. On the other hand, it would in practice be very difficult or impossible to accomplish on account of the safety factors involved in climbing the tree to attach the cable. The initial pull was also slightly downward, to allow for reduction in the height of the attachment point, as it moved round in an arc when the stem bent or pivoted about the roots. A system was devised enabling the angle of deflection of the tree to be measured constantly during the pull by means of distance measurements on the winch rope. Under frozen conditions, the base of the stem was kept free of snow in order to ensure that the soil was frozen to a sufficient depth.

2.2.2. Measurements of the Modulus of Elasticity (MOE)

At the beginning of the pulling of each of a subsample of trees (10 trees of each tree species representing various breast height diameter classes), the strain was recorded at a height of 1.3 m on the stem by means of a Young's modulus sensor while pulling the trees slightly using a small load of 500–600 N in order to cause minimal disturbance to the root–soil plate. This gave a displacement which was very small by comparison with that recorded at failure. The sensor consisted of displacement transducers which enabled the strain of the outer fibres of the stem to be calculated. The length of attachment was measured and recorded in terms of gauge length. The equipment was

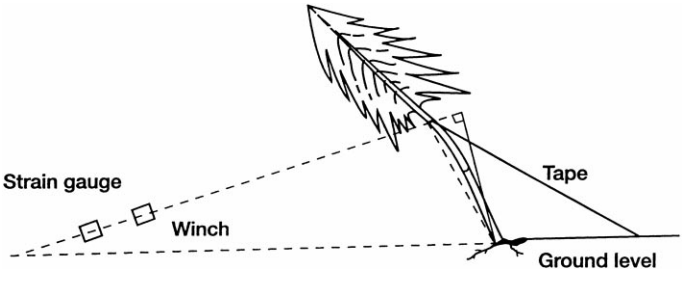


Fig. 1. Layout of tree-pulling system.

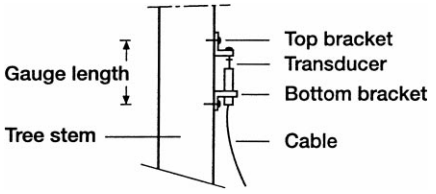


Fig. 2. Setup for the modulus of elasticity measurements.

attached to the tree in the manner indicated in Fig. 2 (see Rogers et al., 1995).

2.2.3. Measurements of tree and soil characteristics

The following stem, crown and root characteristics were measured for all the trees pulled over: tree height, stem diameters at various heights (i.e. 0, 1.3, 6 m), depth and width of crown and root–soil plate (the latter also from broken stems if possible). If the tree broke, the height of the breakage point was also measured.

The root-plate width was taken to be the distance from the centre of the stem to the edge of the central mass of roots and soil, and the root–soil plate depth was correspondingly determined as the maximum depth of the roots (diameter >1 cm). The mean and maximum depths were determined by pushing a measuring rod through the root–soil plate. The only part of the root system measured was the proportion which came out of the ground when the tree was pulled over. The very fine root tips and the long, rope-like surface roots which break off during winching were not measured. The function of these roots was considered to be primarily nutritional. Two measurements of the width of the root system were made at right angles to each other to give a mean value for the maximum root diameter.

The crown envelope was divided into 1-m segments and the width of each segment was determined perpendicularly from the stem to extreme branch tip. Crown height was measured from the leader tip to the lowest live branch (diameter >1 cm). The crown area was based on measurements of crown width made at 1-m intervals and summed over the height of the canopy.

For a sub-sample of trees, the weights of the stem, crown and root–soil plate were also measured using a load optimiser (Kajaani Oy, Finland). The weight of the stem applied to that part that was at least 6 cm in diameter, while the part that was <6 cm in diameter was included in the weight of the crown. Stem sections were also taken at a height of 1.3 m above the base of the stem for further determination of tree age and basic density.

The pattern of stem deflection was recorded while pulling the tree with a rope attached 6 m above the base of the stem. The displacement coordinates being read from a measuring tape attached by one end to the same point on the tree, and the horizontal coordinates by means of another tape extended below a tightly stretched horizontal string aligned in the direction of the pulling rope. The tree was pulled in stages, the force being increased in steps of 0.1 kN. The coordinates were also measured at each stage.

Altogether, 115 trees were pulled over, 51 Scots pines, 33 Norway spruces and 11 birches under unfrozen soil conditions and 20 Scots pines while the soil was frozen. Two modes of failure were observed: uprooting and stem breakage. Uprooting was characterised in this context by lifting of the intact root plate, with the tree then falling under its own weight.

2.3. Data analysis

The maximum resistive moment at the stem base was calculated from the pull applied and the height of attachment of the pulling rope, as follows:

$$\mathrm{BM}_{\max} = F_{\max} \times \mathrm{pull}\ (h) \quad (1)$$

where $\mathrm{BM}_{\max}$ is the maximum applied moment (Nm), $F_{\max}$ the maximum applied force (N) and pull(h) the height of the pull (m). The bending moment due to the weight of the offset stem and crown was neglected, because weight was available only from sample of trees. This results in a small underestimation of the overall bending moment (≈5–20% depending on tree size, see Peltola, 1995), because the figure does not include the gravitational force due to tree's own weight. Thus, the maximum applied bending moment is expected to equal the maximum resistive bending moment.

The modulus of rupture (MOR (Pa)) for the trees which broke was calculated using the relationship between $\mathrm{BM}_{\max}$ (Nm) required to break the stem at breast height, and the diameter at breast height (DBH,

(m)), as follows:

$$\mathrm{MOR} = \frac{32 \times (\mathrm{BM_{max}}/\mathrm{DBH}^3)}{\pi} \quad (2)$$

Since Eq. (2) calculates the bending moment at breast height (1.3 m), the length of the lever arm (cable attachment height) used in the calculation of $\mathrm{BM_{max}}$ (Nm) had to be reduced by 1.3 m to compensate for this.

The modulus of elasticity (E, (Pa)) was obtained as follows (Rogers et al., 1995):

$$E = \frac{(\mathrm{BM} \times y)}{(I \times e)} \quad (3)$$

where BM (Nm) is the applied bending moment at a particular strain gauge level, y the distance from the centre of the tree to the centre of the strain gauge (m), I a second moment of area of the section, which is assumed to be circular (m), and e the strain in that section (m). The bending moment is given by $\mathrm{BM}=FL$ (kNm), where F is the applied load (N), and L the lever arm (m) (equals $6-1.3=4.7$).

The data were analysed statistically using SPSS for Windows. Correlations were first calculated between $\mathrm{BM_{max}}$ and various physical characteristics of trees, such as tree height, DBH, stem weight, root-soil plate depth and width, root-soil plate weight, crown area, and combinations of parameters, such as tree height multiplied by the square of DBH (stem volume) and stem taper (H:DBH, where H is the tree height). This was done separately for each species and failure type. Further correlations were also calculated between various physical characteristics of the trees and the MOR, MOE and angle of stem deflection at $\mathrm{BM_{max}}$. Secondly, stepwise regression was used to obtain the best relationships between the combinations of tree characteristics and $\mathrm{BM_{max}}$ needed to cause uprooting and stem breakage in individual trees.

3. Results

3.1. Mode of failure

Uprooting was most prevalent in the trees winched over when the soil was not frozen, i.e. 78 of the 95 trees were uprooted and the remaining 17 suffered stem breakage. All 20 trees, tested while the soil was frozen, broke (Table 2).

On average, trees with a greater rooting depth had a higher $\mathrm{BM_{max}}$, but root-plate width did not correlate with $\mathrm{BM_{max}}$ in the same way. The various species were fairly similar in the rooting depth and width of the trees, but it must be remembered that the data on each species represent stands of different ages.

The angle of stem deflection at $\mathrm{BM_{max}}$ showed a significant inverse correlation with tree height in the case of the uprooted trees but not the broken ones, except for the pines studied when the soil was frozen. The Scots pines, Norway spruces and birches in unfrozen soil, all provided the maximum resistance on average when the crown was deflected by $\approx 20^\circ$, while the Scots pines in frozen soil were deflected by $\approx 39^\circ$ at $\mathrm{BM_{max}}$.

Trees had obviously the largest stresses in the lower portion of the stem as could be expected on the basis of tree-pulling height used. The average height of stem breakage was 0.34 m for the 17 trees that broke in unfrozen soil, varying in the 0–0.9 m range from the base of the stem for Scots pine, 0–1.1 m range for Norway spruce and 0–0.25 m range for birch, with a range of 0–1.35 m for the Scots pines in frozen soil (average 0.57 m). The ratio of stem failure height to rope attachment height was, therefore, also very low on average (Fig. 3), demonstrating that many of the failures seemed to be associated with the relative height of rope attachment (correlation -0.62, $p<0.01$).

3.2. Maximum resistive bending moment for uprooting and stem breakage

Significant correlations were found between $\mathrm{BM_{max}}$ and tree height (H), DBH, stem weight, taper (H:DBH), rooting depth and crown area ($p<0.01$), with DBH explaining the greatest proportion of the variation in $\mathrm{BM_{max}}$ for all the trees measured. Stem weight was even better correlated with $\mathrm{BM_{max}}$, but was measured only in a sub-sample of trees (see Table 3).

Tree height multiplied by the second power of DBH was found to predict the $\mathrm{BM_{max}}$ for uprooting best under unfrozen soil conditions (Table 4). The data were forced through zero because, as the size of a tree (e.g. in terms of stem weight) approaches zero, so should the bending moment required to uproot or

Table 2
Summary statistics from the Finnish tree-pulling database for uprooted and broken Scots pines, Norway spruces and Birch sp under unfrozen and frozen soil conditions[a]

Variable	Unfrozen soil						Frozen soil
	Scots pine		Norway spruce		Birch		Scots pine
	Uprooted (44)	Broken (7)	Uprooted (26)	Broken (7)	Uprooted (8)	Broken (3)	Broken (20)
Height (m)	14.3	12.5	15.0	16.0	17.3	12.7	12.2
	(2.6)	(1.0)	(2.5)	(2.7)	(2.9)	(2.0)	(1.0)
DBH (cm)	14.9	11.7	15.8	16.7	17.2	11.2	11.1
	(5.3)	(2.8)	(2.9)	(2.7)	(4.0)	(2.5)	(2.1)
H/DBH	102	111	96	98	103	115	112
	(18.5)	(19.1)	(13.7)	(11.9)	(19.6)	(13.7)	(12.8)
Root–soil plate radius (m)	1.2	–	1.3	–	1.4	–	–
	(0.3)	–	(0.5)	–	(0.5)	–	–
Root-soil plate depth (m)	0.4	–	0.4	–	0.4	–	–
	(0.1)	–	(0.1)	–	(0.1)	–	–
Bm_{max} (kNm)	13.5	7.7	13.5	17.5	19.2	5.9	9.0
	(14.5)	(4.0)	(6.8)	(8.0)	(10.7)	(3.6)	(4.7)
Angle of stem at BM_{max}, (degrees)	18.4	33.2	20.1	23.0	19.7	42.0	39.6
	(9.7)	(15.2)	(8.8)	(8.5)	(5.0)	(9.9)	(8.8)
Breakage height (m)	–	0.19	–	0.57	–	0.15	0.57
	–	(0.35)	–	(0.48)	–	(0.13)	(0.34)
Basic density (kg m^{-3})	430	438	401	360	477	452	–
	(54)	(23)	(35)	(52)	(31)	(37)	–

[a] Number of observations are presented relative to each tree species and damage category. Means with standard deviations are presented for each variable (the latter in parenthesis).

break it. The decreasing order of tree species resistance to uprooting was: Scots pine (tap roots in many cases)>birch>Norway spruce. Correspondingly, the third power of DBH explained the BM_{max} for stem breakage best, with a decreasing order of resistance to stem breakage: birch>Scots pine>Norway spruce. Highly tapering trees seemed to be more liable to stem breakage than uprooting (given equal DBH), and

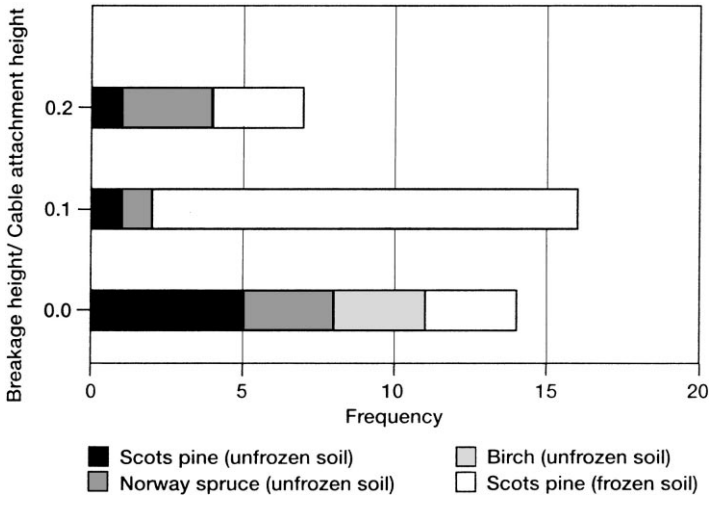


Fig. 3. Histogram for the ratio of failure height to cable attachment height in the broken trees.

Table 3
Correlation between certain physical characteristics of the trees and maximum resistive bending moment

Variable	Scots pine		Norway spruce		Birch		Scots pine Broken
	Uprooted	Broken	Uprooted	Broken	Uprooted	Broken	
Height	0.752**	0.866*	0.634**	0.804*	0.665	0.976	0.809*
DBH	0.932**	0.913**	0.896**	0.889**	0.809*	0.978	0.972**
H/DBH	0.818**	0.861*	0.362	0.043	0.368	0.566	0.904**
Stem weight	0.972**	–	0.974**	0.964**	0.790*	1.00*	–
Root-soil plate depth	0.693**	–	0.495*	–	0.718*	–	–
Root-soil plate radius	0.476*	–	0.606	–	0.661	–	–
Crown area	0.875**	0.788*	0.893**	0.973**	0.827*	0.682	0.801**
Angle of stem at BM_{max}	−0.513**	0.181	−0.406*	−0.414	−0.454	−0.998*	−0.531*

* Pearson correlation is significant at the 0.05 level (2-tailed); **Pearson correlation is significant at the 0.01 level (2-tailed).

larger forces on average were needed both, to uproot and to break them.

The resistance of a Norway spruce to uprooting and stem breakage was always lower than that of the equivalent Scots pine (or birch), reflecting the shallower rooting and/or lower MOR of this species. All the Scots pines tested under frozen soil conditions broke, and a much higher BM_{max} was needed for stem breakage than under unfrozen conditions. The best regressions also show that the BM_{max} required to uproot a tree increases with increasing tree height for a fixed taper or with increasing DBH for a fixed tree height. Because the ability of a tree to resist uprooting is related directly to its stem volume expressed by height·DBH^2, any increase in stem diameter or tree height will increase the resistance of the tree to uprooting.

As the resistance to stem breakage is, correspondingly, a function of DBH cubed. an increase in stem diameter will lead to an increase in the BM_{max} required to break a tree. Note that an increase in height only, e.g. from a tree with a height of 16 m and DBH of 20 cm to one with a height of 20 m and DBH of 20 cm, results in no change in the resistance to stem breakage.

3.3. MOR and MOE

The BM_{max} required to rupture the outer fibres of the stem (and result in breakage) and the DBH were

Table 4
Best regressions between maximum resistive bending moment and various tree characteristics for uprooting and stem breakage in Scots pine, Norway spruce and birch, including variables and slopes with R^2[a]

Damage type	Tree species	Variables[b]	Slope	R^2	SE	*F*	Sig. *F*	N
Unfrozen soil:								
Uprooting								
	Scots pine	$H\times DBH^2$	3.589	0.967	3628	1258	0.0001	44
	Norway spruce	$H\times DBH^2$	3.315	0.946	3573	440	0.0001	26
	Birch	$H\times DBH^2$	3.320	0.917	6657	77	0.0001	8
Stem breakage								
	Scots pine	DBH^3	3.789	0.926	2487	76	0.0001	7
	Norway spruce	DBH^3	3.434	0.954	4405	124	0.0001	7
	Birch	DBH^3	3.798	0.997	412	767	0.001	3
Frozen soil: *Stem breakage*								
	Scots pine	DBH3	5.760	0.988	1117	1611	0.0001	20

[a] S.E., standard errors; *F*, *F* values, and Sig. *F*, significance of regression models developed;and *N* number of observations.
[b] *H*, tree height (m); DBH, diameter at breast height (cm).

Table 5
Mean values and standard deviations (S.D.) of MOE (GPa) and MOR (MPa) for Scots pine, Norway spruce and birch

Variable		Tree species		
		Scots pine	Norway spruce	Birch
In unfrozen soil				
MOE	mean	11.35	7.73	11.06
	S.D.	3.51	1.94	5.46
	N[a]	10	8	9
MOR	mean	37.29	36.26	40.68
	S.D.	7.71	6.99	5.23
	N	6	7	3
In frozen soil				
MOR	mean	63.33	–	–
	S.D.	9.16	–	–
	N	20	–	–

[a] Number of observations.

used to calculate the MOR for each tree that broke (see Table 5). The average MOR values were in decreasing order of tree species: birch>Scots pine>Norway spruce, but the average differences between the species were not large.

Correspondingly, the average modulus of elasticity (MOE) values were in a decreasing order of the tree species: Scots pine>birch>Norway spruce. However, there was significant variation in MOE between the trees within one species, as can be seen from the standard deviations. Nevertheless, MOE did correlate negatively with wood density for both failure types. In addition, a close correlation was detected between MOE and the BM_{max} (0.722 for uprooting; 0.914 for stem breakage) and stem taper (0.704 and 0.588, respectively).

4. Discussion and conclusions

4.1. Mode of failure

The most common outcome of the pulling of trees, when the soil was unfrozen, was uprooting, i.e. 78 of the 95 trees suffered this fate and the remaining 17 broke. Under frozen conditions, all 20 trees broke. This indicates that root anchorage was less pronounced relative to stem strength in most cases under unfrozen conditions, whereas the frozen soil provided increased stability of root anchorage. Also Coutts (1986), for example, found for Sitka spruce (*Picea sitchensis*) that the resistance to uprooting was considerably smaller than that required to break the stem, and that there appeared to be scope for strengthening the anchorage of the more weakly anchored trees before there was any risk of extensive stem breakage. In fact, all the evidence from extensive tree pullings suggests that there is a rough balance between uprooting and breaking only for well-drained soils, where rooting is not restricted.

In this study, we found that the trees with a greater rooting depth had a greater average BM_{max}, whereas the root-plate width did not correlate with BM_{max} in this manner. Also Coutts (1986) and Ray and Nicoll (1998) similarly suggested that, as the strength of the root anchorage influences the mode of failure, a positive relationship could be expected between this, quantified in terms of BM_{max}, and rooting depth of Sitka spruce trees (*Picea sitchensis*). Likewise, they failed to demonstrate any significant relationship between BM_{max} and root-plate diameter. On the other hand, Moore (1999) found a significant relationship both between BM_{max} and root-plate diameter and rooting depth for Radiata pine (*Pinus radiata*).

Coutts (1986) also identified the weight of the root–soil plate as the second largest component of root anchorage. We also measured this in a sub-sample of trees, but it was in practice very difficult to do so even roughly (in most cases, there was soil dropping off the root–soil plate). In addition, measurements of root–

soil plate depth and width gave only rough estimates, because many roots broke during pulling and the exact dimensions were difficult to measure (it would, therefore, be unrealistic to include these as dependent variables in models).

The angle of stem deflection at BM_{max} showed a significant inverse correlation with tree height for the uprooted trees, but not for the broken ones (except for the broken pines standing in frozen soil). Also Morgan and Cannell (1987) have argued that, because the bending moment applied to the tree is linearly related to its size, the angle of stem deflection will be smaller for larger trees. This theory applies correctly, however, if the deflection is approximately $<14^{\circ}$. In this study, Scots pines, Norway spruces and birches provided their maximum average resistance when the crown was deflected by $\approx 20^{\circ}$ under unfrozen conditions. The average stem deflection observed here at BM_{max} is somewhat higher than that (about 10°) presented by Somerville (1979) for Radiata pine, Oliver and Mayhead (1974) for Scots pine and Fraser and Gardiner (1967) for Sitka spruce on well-drained soils (such as brown-earths or podzols). However, the tape measurements might overestimate the deflection to some extent, because the effect of the curve of stem bending was neglected in this study.

All the trees broke at a point equivalent to <10% of the tree height. It demonstrates that, obviously, the largest stresses were occurring in the lower portion of the stem, as could be expected because the relative height of rope attachment was between one-third and one-half of the tree height (see Metzger, 1893; Wood, 1995).

4.2. Maximum resistive bending moment for uprooting and stem breakage

In this study, we observed that a tree with a larger DBH for the same height will tend to resist breakage and uprooting better, on average, than a tree with a smaller DBH. Highly tapering trees (low height:DBH ratio) seemed to be more liable to stem breakage than uprooting on average (for an equal DBH), i.e. a larger BM_{max} was needed for uprooting. Furthermore, the resistance of a Norway spruce to uprooting and stem breakage can always be expected to be lower than that of the equivalent Scots pine (or birch), reflecting the shallower rooting and/or lower stem resistance of this species, which is in line with earlier findings reported in the literature (Kalela, 1949; Hakkila, 1972; Lavers, 1969). Under frozen soil conditions, all the Scots pines broke and a much higher BM_{max} was needed for stem breakage than in unfrozen soil, indicating increasing root anchorage due to soil frost and also an increase in stem stiffness (the larger MOE of frozen wood).

Significant relationships between the BM_{max} of a tree and various measures of its size have been proposed earlier by Fraser (1962) and Fraser and Gardiner (1967), for example, who found a linear relationship between BM_{max} and stem weight for Sitka spruce. Likewise, Frederickson et al. (1993) and Moore (1999) found significant relationships between BM_{max} and total tree weight, stem volume, DBH and tree height for Loblolly pine (*Pinus ponderosa*) and Radiata pine, respectively. In this study, the relationship between BM_{max} and stem volume index, calculated as $DBH^2 \times$height was a particularly clear one ($r^2=0.94$), as was also found here in the case of the uprooted trees. Papesch et al. (1997) similarly found quite a close correlation between DBH and BM_{max} ($r^2=0.76$) for Radiata pine, whereas the correlation between tree height and BM_{max} was weaker ($r^2=0.59$), and this also resulted in a weaker correlation between stem volume and BM_{max}.

4.3. MOR and MOE

The average MOR values of the tree species showed a decreasing order of species: birch>Scots pine>Norway spruce. The differences between the species were, in general, not large, but MOR was significantly lower on average than the values reported for clear wood specimens (green wood). Lavers (1969), for example, suggested typical values of 46 MPa for Scots pine, 36 MPa for Norway spruce and 63 MPa for birch. Comparison with the values, measured indirectly here, for MOR reveals that large differences in MOR exist between knot-free wood and whole trees. The MOR measured for Scots pine, for example, was on average 85% of the typical value for knot-free wood. The effect of branches and other stem defects on the overall strength of the stem (reduced wood strength) coupled with the violation of the assumption of a uniform stress distribution (see Metzger, 1893; Wood, 1995) due to the height of cable attachment has led to

differences between the results of winching experiments and those of theoretical calculations.

Huge variations in MOE also existed between the trees (4.5–20 GPa), and the differences between the species were not as expected. Lavers (1969), for example, suggested typical MOE values of 7.3 GPa for Scots pine, 6.3 GPa for Norway spruce and 9.9 GPa for birch. Again, these results reveal a large difference between the MOE for knot-free wood and the values for whole trees. Unfortunately, the data were too limited to allow any overall conclusions to be reached. Furthermore, the MOE and MOR values tend to be underestimated by ignoring the effect of the overhanging mass.

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